



Mriganka Dey

PRODUCT ENGINEER

Healthcare RCM & Automation Systems

"The best interface is the one that disappears the work just gets done."

- Design philosophy

CONTACT

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PROFILES

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🌐 mrigankad

AUTOMATION & RPA

Desktop UI Automation

WinAppDriver Appium

pyjab / JAB n8n

Playwright

LANGUAGES

Bengali BD

Hindi IN

English GB

EDUCATION

Sikkim Manipal Institute of Technology

B.Tech, CS & Engineering · 2025

Google UX Design Certificate

Coursera · 2023

Microsoft UX Professional Certificate

Coursera · 2024

ENGINEERING

Python React TypeScript

JavaScript Next.js Go

Java SQL FastAPI

Node.js Tailwind CSS Vite

Docker AWS Git

REST APIs

AI & DATA

Multi-Agent Systems RAG

Agent Orchestration

Text-to-SQL RPA

Prompt Engineering VLMs

Gemini API Groq

Auto-Labeling Power BI

Qlik Sense Tableau

BI Migration TensorRT

About Me

I build the automation layer that lets healthcare get paid faster turning slow, manual revenue-cycle work into systems that mostly run themselves.

I'm a Product Engineer who moved from designing interfaces to engineering the systems behind them full-stack products, desktop & workflow automation, analytics tooling, and multi-agent AI. My work has consistently lived where **data-heavy operations meet automation**: billing engines, reporting migrations, and bots that drive legacy desktop applications no API can reach.

My aspiration is to bring that stack into **healthcare Revenue Cycle Management** the "touchless" revenue cycle. I want to build agentic and RPA-driven workflows for eligibility, claims, denial prediction, and payment posting that cut cost-to-collect and give clinical teams back their time. I learn a domain deeply, then engineer the product that makes it disappear into the background.

What Sets Me Apart



Designer + Engineer in one

Google & Microsoft UX-certified and a shipping full-stack engineer I design the product and build it, so RCM workflows are both usable and technically sound.



Low-level depth to agentic breadth

From a 3D engine written from scratch in Go to multi-agent AI frameworks I reason across the whole stack and pick the right tool for touchless revenue-cycle workflows.



Automation where APIs can't reach

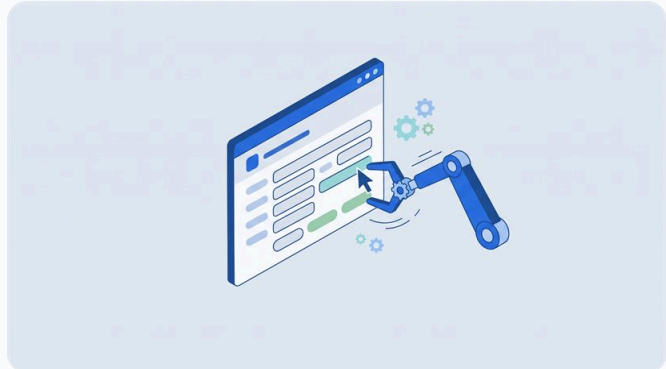
Desktop RPA on legacy apps via WinAppDriver & Java Access Bridge proven with GE Healthcare's CSCS tool, exactly the automation legacy EHR/PM systems demand.



Ships end-to-end, solo

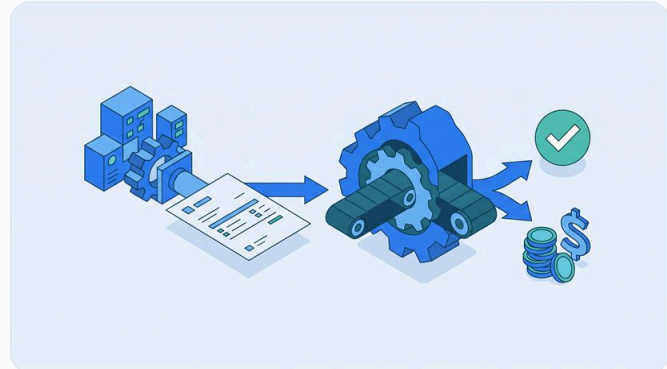
Backend, frontend, SDK, and deploy complete products delivered independently, plus merged contributions to major open source (three.js, vanna).

Healthcare & RCM Focus



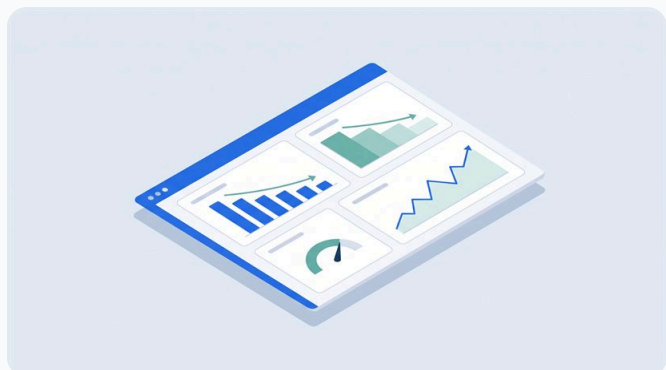
RPA for Legacy EHR/PM Systems

Desktop UI automation (WinAppDriver, Appium, pyjab/JAB) applied to the exact class of legacy practice-management and EHR desktop apps RCM teams automate for eligibility checks and status follow-up.



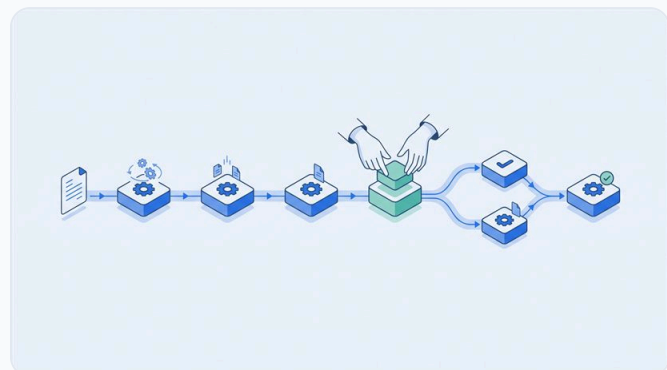
Claims & Billing Engines

Built a multi-tenant SaaS invoice/billing engine with configuration-driven workflows directly transferable to claims generation, remittance handling, and payer-specific billing logic.



Denial & AR Analytics

Text-to-SQL analytics and BI dashboards (Power BI, Qlik, Tableau) suited to denial-rate tracking, AR aging, and cost-to-collect reporting that make the revenue cycle a board-visible control system.



Agentic / Touchless Workflows

Multi-agent orchestration with persistent memory and exception handling the architecture behind autonomous prior-auth, claim-submission, and denial-appeal pipelines that escalate only edge cases.

Experience

Jun 2026 - Present · Remote

Freelance Product Engineer

Independent

- Partner with healthcare and SaaS companies to make their billing and revenue-cycle operations increasingly agentic designing autonomous, exception-driven workflows that remove manual touchpoints from the revenue cycle.
- Working with Medisoft LLP to transform their billing cycle into an agentic pipeline automating claim/invoice generation, follow-up, and reconciliation so staff intervene only on exceptions.
- Build RPA and multi-agent automation on top of existing tools, embedding AI-driven decisioning into established billing and revenue-cycle workflows to cut cost-to-collect.

freelance

Oct 2025 - Jun 2026 · Bengaluru

Product Engineer

Joules To Watts Pvt Ltd

- Partnered with GE Healthcare to automate test cases for their CSCS tool used in patient health monitoring building RPA against a legacy clinical desktop application, the same EHR/PM-style automation central to healthcare RCM.
- Owned product design and full-stack frontend delivery for enterprise SaaS platforms across talent, pricing, billing, and analytics the same data-heavy, workflow-driven surface that defines RCM products.
- Built the J2W Invoice Engine, a multi-tenant SaaS billing platform with dynamic theming and configuration-driven billing logic architecturally analogous to claims and payer-specific billing systems.
- Developed the React/TypeScript frontend for Talent Sphere, delivering role-based dashboards and analytics-driven workflows for operational decision-making.
- Established design-system standards and frontend architecture improving consistency and scalability across all internal platforms.

Joules to Watts

Jan 2025 - Jul 2025 · Thane

Product Engineer Intern

Mahindra Teqo

- Designed and built a real-time performance analytics dashboard with KPI visualization and automated reporting for monitoring operational deviations a pattern that maps cleanly to denial and AR monitoring.
- Improved planning workflows in the PV Lab Scheduler with route visualization and interactive scheduling experiences.
- Led the UI redesign of TeqoNoc V2, modernizing a complex enterprise monitoring platform for long-term scalability.

teqo

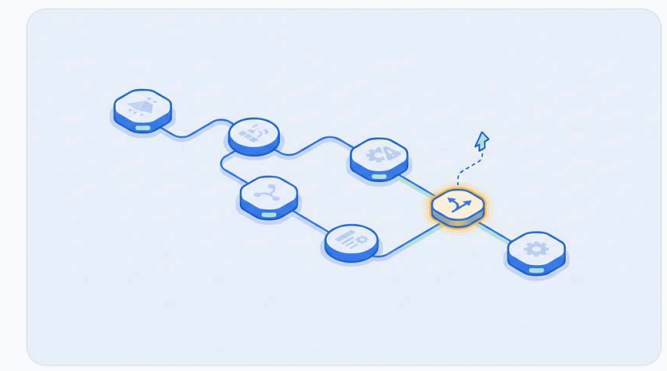
Selected Projects



GenBI

Natural-language-to-SQL analytics engine generating queries, charts, and BI outputs the reporting core for denial-rate, AR-aging, and cost-to-collect dashboards.

Text-to-SQL Analytics



Novel-OS

Production-grade multi-agent framework with persistent memory and a 5-stage pipeline the orchestration blueprint for agentic, touchless claims and denial workflows.

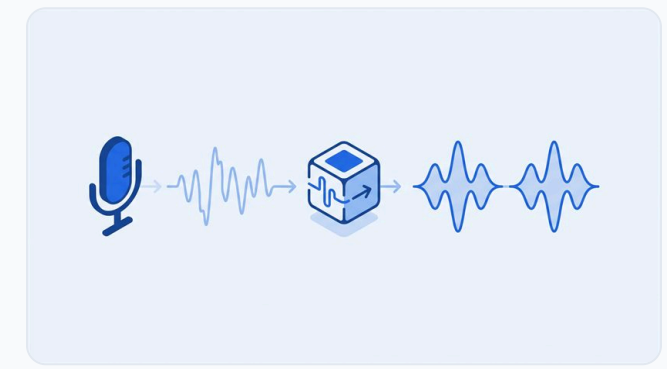
Multi-Agent RAG



Desktop Automation Suite

Windows application automation with WinAppDriver, Appium, and Java Access Bridge to drive legacy desktop apps end-to-end the exact RPA approach used against EHR/PM systems.

RPA WinAppDriver

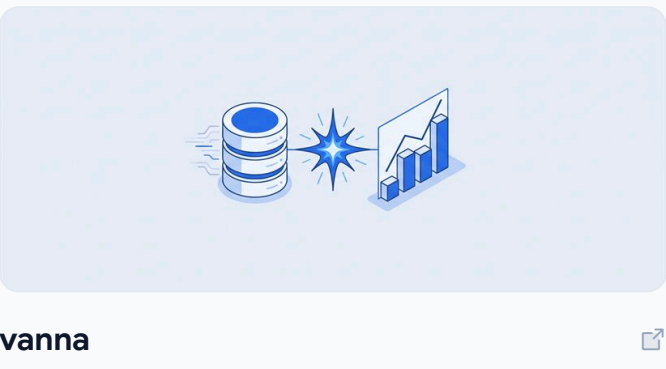


Mimicry

Self-hosted zero-shot voice-cloning platform with a FastAPI backend, management frontend, and async Python SDK shipped as a complete deployable full-stack product.

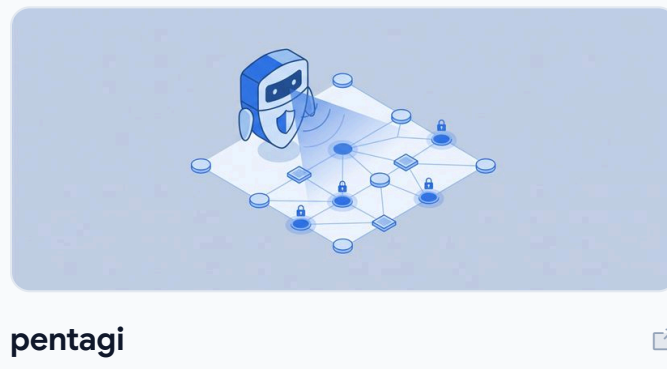
Full-Stack AI FastAPI

Open Source Contributions



vanna

Resolved Gemini & FastAPI issues in an AI data-analytics assistant.



pentagi

Security & reliability fixes for an autonomous agentic workflow system.